

# Python Area Calculator

## Master Code — While Loop, Conditions, No User-Defined Functions

```
while True:

    print("\n===== AREA CALCULATOR =====")
    print("1. Area of Circle")
    print("2. Area of Rectangle")
    print("3. Area of Square")
    print("4. Exit")
    print("=====")

    try:
        choice = int(input("Enter your choice: "))

        if choice == 1:

            try:
                radius = float(input("Enter the radius: "))

                if radius > 0:
                    area = int(3.14159 * radius * radius)
                    print("Area of Circle =", area)
                else:
                    print("Radius must be greater than 0.")

            except ValueError:
                print("Invalid input! Please enter a number.")

        elif choice == 2:

            try:
                length = float(input("Enter the length: "))
                breadth = float(input("Enter the breadth: "))

                if length > 0 and breadth > 0:
                    area = int(length * breadth)
                    print("Area of Rectangle =", area)
                else:
                    print("Length and breadth must be greater than 0.")

            except ValueError:
                print("Invalid input! Please enter a number.")

        elif choice == 3:

            try:
                side = float(input("Enter the side: "))

                if side > 0:
                    area = int(side * side)
                    print("Area of Square =", area)
                else:
                    print("Side must be greater than 0.")

            except ValueError:
                print("Invalid input! Please enter a number.")

        elif choice == 4:
            print("Program ended.")
            break

        else:
            print("Invalid choice! Please enter 1, 2, 3, or 4.")
```

```
except ValueError:  
    print("Invalid choice! Please enter a number.")
```

## Choice 1 — Area of Circle

Integer Input

Enter your choice: 1

Enter the radius: 5

Area of Circle = 78

Decimal Input

Enter your choice: 1

Enter the radius: 5.5

Area of Circle = 95

Scientific Notation Input

Enter your choice: 1

Enter the radius: 5e0

Area of Circle = 78

## Choice 2 — Area of Rectangle

Integer Input

Enter your choice: 2

Enter the length: 10

Enter the breadth: 5

Area of Rectangle = 50

Decimal Input

Enter your choice: 2

Enter the length: 10.5

Enter the breadth: 5.5

Area of Rectangle = 57

Scientific Notation Input

Enter your choice: 2

Enter the length: 1e1

Enter the breadth: 5e0

Area of Rectangle = 50

## Choice 3 — Area of Square

Integer Input

Enter your choice: 3

Enter the side: 6

Area of Square = 36

Decimal Input

Enter your choice: 3

Enter the side: 6.5

Area of Square = 42

Scientific Notation Input

Enter your choice: 3

Enter the side: 6e0

Area of Square = 36

## Invalid Input Handling

Invalid Input Example

```
Enter your choice: 1
Enter the radius: abc
Invalid input! Please enter a number.
```

Note: Python input is initially received as text. Numeric integer, decimal, and scientific-notation inputs are converted to float for calculation. Every calculated area is converted using `int()`, so the displayed result is an integer.